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Modeling pedestrians' road crossing behavior in traffic system micro-simulation in China

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Abstract

In many Chinese cities, pedestrian's road crossing behavior is different from that of pedestrians in developed countries. This paper presents a pedestrian model for traffic system micro-simulation in China. Considering the high rate of signal non-compliance, we classify pedestrians into two types: law-obeying ones and opportunistic ones. Opportunistic ones decide whether to violate traffic signal during red man, depending on the states of some external factors (like policeman, vehicle flow and other pedestrians' behaviors). Questionnaires were used to determine the proportions of these two types of pedestrians under different circumstances. In addition, a time gap distribution extracted from videotape were used to determine the criterion for pedestrians to decide whether to walk or wait when they conflict with vehicle flows. However, simulation results deviate from the data extracted from videotape in some degree. By adjusting the parameters on the basis of analyzing the occurrence of the deviations, the simulation results agree with the field results better. This model has represented the high rate of pedestrians' red light running and the mixed characteristics of traffic flows in Chinese cities, and it may be applicable in the micro-simulation of traffic system in other developing cities.

Previous Next

Keywords

Microscopic simulation; Modelling; Pedestrian; Traffic violation

1. Introduction

Traffic simulation has been widely used to evaluate control scenarios in recent years. Pedestrian is a so important entity in traffic flows that it could not be ignored when we lay a managing course. Nowadays, a number of studies have been conducted on pedestrian simulation.

As for the microscopic aspect, several kinds of simulation model have been developed widely. Benefit Cost Cellular Model, Magnetic Force Model, and Social Force Model have been developed for a period of time. These models mainly concern

explanation of the phenomena of pedestrians' queuing, way finding and self-organization (Teknomo et al., 2000). In addition, cellular automata model has been introduced to provide a microscopic description of pedestrians. For instance, bi-directional pedestrian movement was modeled based on cellular automata by Blue and Adler, 2001, Fang et al., 2003.

However, little information has been published concerning micro-simulation of pedestrians' signal non-compliance decision when they are crossing road. Researches on pedestrians' violation mainly focus on sociology and psychology to find out which situational factors could have influence on pedestrians, and whether it would be different with different age or gender.

A questionnaire was distributed to 205 Israeli college students. Respondents were asked to answer how the presence of a certain factor, like traffic volume, weather, other pedestrians' violation, would affect their behaviors as crossings with a 'don't walk' sign (Yagil, 2000). Hamed (2001) paid attention to the divided streets (with refuge island). Impact of the pedestrian waiting countdown timer on pedestrian behavior was also studied (Keegan and Mahony, 2003).

Liu et al. (2000) developed a micro-simulation tool for assessing pedestrian delays and traffic management. They classified pedestrians into two types: "law-obeying" ones and "opportunistic" ones. But proportion of each type was simply predetermined before simulation, not depended on the decision results of 'pedestrians' and varies along with the traffic conditions.

As a developing country, China has her own traffic characteristics. Researches show the proportion of walking in the total egression comes to 40% in big cities in China, and it must be higher in smaller cities or in towns. Owing to the weakness of safety consciousness and misgoverning, pedestrians' violation in China is popular. Report on the Investigation of Pedestrians' Road-crossing Behavior in the Main Road of Guangzhou in 2003 shows, in average, 23.3% of pedestrians took an illegal action while crossing streets, and the rate came to 66.7% in certain sections. Most pedestrians always turn a blind eye to the traffic lights. Consequently, the road is often under the situation that vehicles and pedestrians being influenced by each other and the mixed traffic flow has been a typical feature of the Chinese city traffic.

Due to the special features of Chinese city traffic, application of the micro-simulation tools developed by foreign countries in China has not obtained obvious effect. He (2003) once studied the applicability of CORSIM and SIMTRAFFIC for signalized intersections in Beijing, China, results show that their vehicle models can achieve favorable effect, but the pedestrian and bicycle models could not truly describe the situations at intersections. Modeling of pedestrians' behavior and its impact on traffic flow is a key point need to be solved in the field of traffic system micro-simulation in our country.

Some researchers in China have realized this particularity, and correlative work is being done, but it still needs a long way to go. Zhang et al. (2004) used cellular automata to derive a two-dimensional model for simulation of pedestrian's crossing dynamics. Simulation of a crosswalk without signal was carried out. Interactions between pedestrians and vehicles were described in some degree, but pedestrians' fighting for right of way with vehicle drivers has not been fully presented. Gao et al. (2004) regarded pedestrians as a disturbance to vehicle flows. They observed a pedestrians' crossing point in the city of Wuhan and brought forward a pedestrians' disturbance model for traffic micro-simulation based on Poisson distribution. Parameters were calibrated by field data. However, pedestrians' crossing choices are seriously impacted by the current traffic conditions, especially the vehicle gaps. Gao et al. have not taken these factors into consideration.

In simulation, it is an important thing to let the 'pedestrian' know how to pass through the vehicle flow safely after their violation decisions. Many researchers have done investigations on this problem. In general, Gap Acceptance is widely used as the judging criterion of pedestrians' safety crossing. Each pedestrian is supposed to have a critical gap, which is calculated by this equation (Wan and Rouphail, 2004)

$$\mathrm{CG} = \frac{L}{S} + F \tag{1}$$

where L is the crosswalk length, S is the pedestrian's average walking speed. F is a safety factor reflects pedestrian aggressiveness. Pedestrians will do comparison between the vehicle gaps and the critical gap. The gap, which is smaller than the critical gap, will be rejected. In 1991, Hunt and Griffiths took two-direction vehicle flows into account and gave probability matrices for rejection of crossing opportunities for 10.0m wide roads (Lyons et al., 2001).

Appearance of the traffic jam in rush hours and the increase of accident rate have a strong relation to the participators' non-compliance. In order to truly reappear the traffic condition of the signalized intersections in China, it is necessary to develop a

model basing on human behavior for simulating pedestrians' red light running when crossing roads.

This paper will give a description of a model establishment for simulating pedestrians' road crossing behaviors in Chinese cities. In this model, pedestrians will completely make a decision whether run through red light by them. Model parameters will be calibrated by the analysis of a questionnaire investigation of pedestrian violation behavior at signalized intersections in the city of Xi'an, China, along with the videotape data. Comparison between simulation results and data extracted from videotape is given in the end.

2. Model framework

Nowadays, most intersections have equipped with a 'walk/don't walk' sign for pedestrians in Chinese cities. Nevertheless, since the lack for safety consciousness and effective punishment, a majority of pedestrians do not comply with the traffic lights. Red light running is popular. Once a proper gap occurs, pedestrians will go through immediately. Furthermore, interactions among pedestrians themselves are serious. When someone successfully cut in the gap between two vehicles, others will speed up to follow him/her and make the latter vehicles yield to them. Since Road Traffic Safety Law was implemented on May 1st 2004, punishment for pedestrians' violation has been aggravated. Consequently, proportion of pedestrians' violation of a certain road section may change with the strictness of the government of policeman.

We classified pedestrians into two types, according to their behavior during red light:

I. People who comply with the traffic law, and do not take an illegal action. Most elders belong to this type.

II. People who look for appropriate gaps between vehicles to cross during red man. Most pedestrians in China could be considered as this type.

We hypothesize that if external factors, like policeman and other pedestrians' behavior, have no effect on a pedestrian, his/her behavior during the red man will be relatively steady, and it only depends on his/her age, morality and safety consciousness. In a large quantity of people, proportions of these two types of pedestrians are constants.

In real life, human's decision-making and action process is very complex, and it is impacted and restricted by many factors. Pedestrians' behaviors may vary with external factors. For instance, some pedestrians of Type II may become pedestrians of Type I, since they are afraid of being punished by the policeman. Some pedestrians of Type I may follow other pedestrians to jump the red light and become a member of pedestrians of Type II. Due to the influences of all these external factors, traffic system becomes more complex, unpredictable, with strong randomness. If we only set the proportion of each pedestrian type before simulation, these features could not be represented adequately.

According to the situation in Chinese cities, we finally choose policeman, behavior of other pedestrians and vehicle flow from all the external situational factors, while others will be ignored.

Fig. 1 shows the model framework. Pedestrians are generated according to the proportion of the two types of pedestrians in general conditions. Denote r_1 as the proportion of pedestrians of Type I. Considering the impact of population fluidity and strictness of execution of law, this proportion can be adjusted according to object of simulation. Pedestrians of Type I will stop and wait for the green man when facing the red light, while pedestrians of Type II will look for a proper gap of vehicles to cross through. A number of pedestrians of Type I will follow other people to violate traffic signal if someone else jumps the red light while his/her waiting. We denote this proportion as r_2.

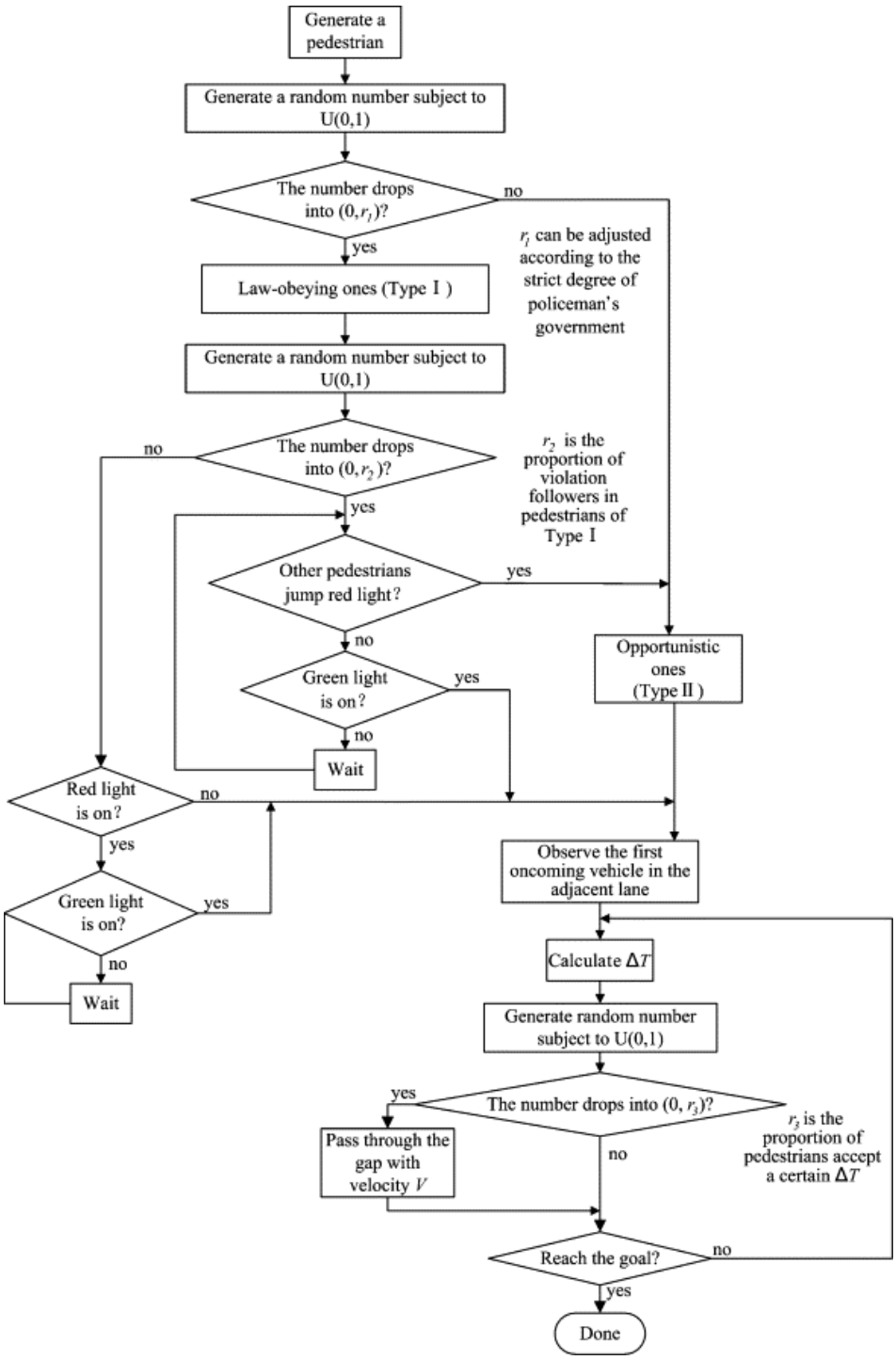

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Fig. 1. Model framework.

In our model, pedestrian takes a "one by one" mode to cross through vehicle lanes. At this time, pedestrian only considers the time gaps between vehicles in the adjacent lane, and he/she would not care what happens in the other lanes until reaching the end point of the adjacent lane. Here, define ΔT as the safety criterion of gap acceptance for pedestrians. Fig. 2 shows the method of calculating ΔT. When a pedestrian walk to the curbside, he/she will estimate the time T_1, which is needed for the first oncoming vehicle in adjacent lane to arrive at his/her position. Meanwhile, T_2, which is the time needed for the pedestrian to reach the safe position (the ultimate position where the pedestrian can just avoid collision with the oncoming vehicle), will be estimated by the pedestrian.

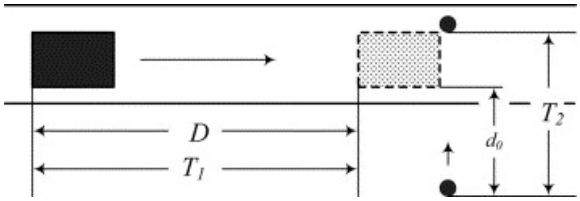

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Fig. 2. Calculation of ΔT.

ΔT is the difference between T_1 and T_2, which can be calculated by Eq. (2)

$$\Delta T = T_1 - T_2 = D/V - (w + d_0)/V_{\text{ped}} \tag{2}$$

Here, D is the distance between the front edge of the oncoming vehicle and the pedestrian, w is the width of the coming vehicle, d_0 is the distance between the position where the pedestrian stands and the vehicle's lateral edge which is near the pedestrian (see Fig. 2). Considering that a pedestrian cannot exactly predict the future action of the vehicle, we use the pedestrian's estimation of the vehicle's instantaneous speed V to calculate T_1, V_{ped} is the average walking speed of pedestrians: 1.2 m/s.

In order to obtain the proportion (r_3) of pedestrians who accept a certain ΔT, we use image-processing software to extract large amount of samples from videotape. It will be introduced in the next section.

3. Parameter calibrations

From the model framework we can see there are three proportion parameters, r_1, r_2 and r_3, need to be calibrated. Questionnaires and image processing technology are used to achieve this goal.

3.1. Questionnaires

In order to find out characteristics of pedestrians' behavior at intersections in Xi'an, questionnaires were distributed from November 22nd to 23rd, 2003. Respondents were 200 students from Xi'an Jiaotong University and 300 pedestrians who were chosen randomly and were willing to be investigated at the intersection which lies on the northeast corner of Xi'an Jiaotong University (the conjunction of West Xianning Road and Xingqing Road). Four hundred and ninety five valid answer sheets were reclaimed. 304 (61.4%) were male and 191 (38.6%) were female. Table 1 shows the age distribution of these respondents.

Table 1. Age distribution of the questionnaire respondents

Age	<18	18–25	26–35	36–45	46–55	>56	Total
Count	19	338	79	25	26	8	495
Proportion (%)	3.8	68.3	16.0	5.1	5.2	1.6	100

The questions mainly focused on pedestrians' behavior at intersections and the degree of the external situational factors (like policeman, traffic volume, weather, and other pedestrians' behavior) work on probabilities of pedestrians' signal non-compliance. Questions, alternative answers and statistic results of the answers are listed in Table 2, Table 3. Some questions may be skipped over by respondents, and these items were treated as invalid ones that had not been counted in the percent calculating for the final results. In addition, there was an alternative answer "hard to say" in the question serials of the probabilities of pedestrians' signal non-compliance under specific situations. If a pedestrian had chosen this item, the corresponding question was also treated as invalid, and it had no effect on the final results.

Table 2. Pedestrians' behavior at intersections

Questions	Answers				
	Always	Often	Sometimes	Never	Missing
Do you walk on the zebra crossing when you cross road?	177	261	54	2	1
	36%	53%	11%	0%	–
Do you look at the state of traffic lights when you cross road?	279	171	38	6	1
	56%	35%	8%	1%	–

Table 3. Probabilities of pedestrians' signal non-compliance under specific situations

Situations	Probability						
	Very high	High	Medium	Low	Very low	Hard to say	Missing
General conditions	6	15	64	141	268	6	1
	1%	3%	13%	29%	55%	–	–
In a hurry	28	125	126	110	87	18	1
	6%	26%	27%	23%	18%	–	–
Long duration of red light	22	72	140	107	125	27	2
	5%	15%	30%	23%	27%	–	–
Presence of other pedestrians who violate traffic signal	38	143	127	84	83	20	0
	8%	30%	27%	18%	17%	–	–
All other pedestrians comply with the traffic signal	5	14	61	133	256	25	1
	1%	3%	13%	28%	55%	–	–
Policeman is on duty at the intersection	7	16	84	93	279	14	2
	2%	3%	18%	19%	58%	–	–
High traffic volume	5	52	38	70	327	3	0
	1%	10%	8%	14%	67%	–	–
Low traffic volume	26	104	126	121	104	13	1
	5%	22%	26%	25%	22%	–	–
Hot day	9	50	117	92	201	26	0

Situations	Probability						
	Very high	High	Medium	Low	Very low	Hard to say	Missing
	2%	10%	25%	20%	43%	–	–
Snowy/rainy day	7	36	67	117	254	14	0
	2%	7%	14%	24%	53%	–	–

From these tables, we can see that more than half of the pedestrians do not have strong safety consciousness while crossing roads. Some pedestrians even never care the state of traffic lights; say nothing of complying with them. Among all the factors listed above, presence of other pedestrians who violate traffic signal, in a hurry, low traffic volume and long duration of red light contributed more to the probabilities of pedestrians' signal non-compliance. What is noticeable and understandable is that presence of policeman did not have distinct impact on pedestrians. This may due to that, policemen had already spent much energy on the guidance of vehicles, and their supervision towards pedestrians seemed to be powerless at that time.

Considering the pedestrians who chose the item "Minimum" as the one who were not going to violate traffic signal, in general conditions, this kind of pedestrians accounted for 55% (we use it as r_1), while under the circumstance that other pedestrians have violated traffic signal, this proportion reduced to 17%. Then, r_2 is calculated by this equation: r_2=38/55×100%=69%, it means that 69% of the pedestrians of Type I may follow others to violate traffic signal. Since this part of pedestrians of Type I only become a red light jumper when other pedestrians violate traffic signal, the final proportion of red light running will fluctuate between 45% and 83%. In this way, differences among various intersections will be represented.

3.2. Extracting ΔT from videotape

Calibrating parameters by the data extracted from videotape is a common method used in traffic simulation research field. Das et al. (2005) collected videotape data of three signalized intersections in New Delhi and obtained pedestrians' critical gap distribution of these intersections. They established an ordered probit model of gap acceptance, which takes the configuration of the intersection, pedestrians' crossing point and attributes of the pedestrian (such as age and gender) into account. Maximum likelihood method was used to estimate the model parameter. However, the vehicle gaps are obtained by the difference of the time when the vehicle passed the position where pedestrian stands. In fact, pedestrians could not exactly predict vehicle's future actions at the time when they decide whether to violate traffic signal. For this problem, we will use the pedestrian's estimation of the vehicle's velocity to calculate the time that needed for the vehicle to reach the pedestrian's position. This is the difference between Das and Manski's method and that of ours. However, pedestrians' estimation of the vehicle's velocity couldn't be extracted from the videotape, so we use the vehicle's instantaneous speed to describe this velocity.

From Fig. 1 we can see pedestrians are supposed to be impacted by nothing but oncoming vehicles while they passing through the gaps between vehicles after their signal non-compliance decisions. In order to obtain pedestrians' real behavior in this condition, we chose a zebra crossing without traffic lights, which is outside the Southeast Gate of Xi'an Jiaotong University, as screening objective. Video camera was placed on the 27th floor of Ruisen Building (a building near the Southeast Gate of Xi'an Jiaotong University). Fig. 3 shows a picture intercepted from the videotape. The zebra crossing lies in the middle. Areas lined out by white rectangles are defined as pedestrians' waiting areas, since most of pedestrians in our country will stand in these areas to wait for proper vehicle gaps to pass through. A pedestrian has to walk through four lanes until she/he reach the non-motor area on the opposite side.

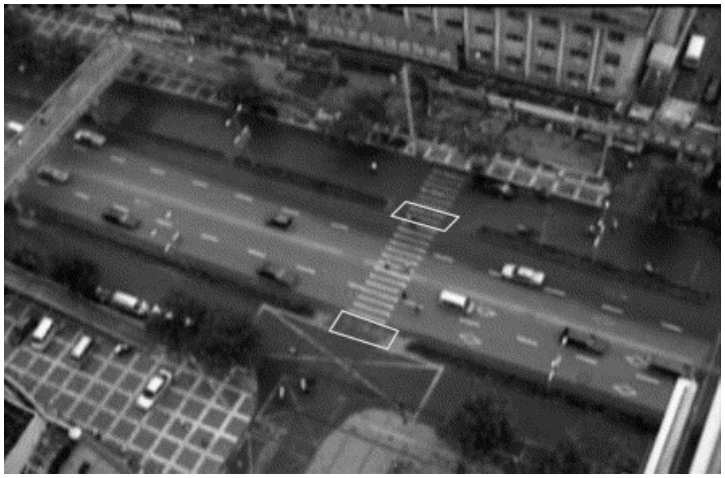

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Fig. 3. Zebra crossing outside the Southeast Gate of Xi'an Jiaotong University.

At first, the video was saved as several files with fmpg format. Then, we used the continuously photographing function of a software tool named Hero3000 to get a serial of pictures. A video with the length of 1 s can obtain 24 pictures, which have the same time interval.

A program was written to extract ΔT from the picture serials by us. From the time a pedestrian happen to arrive at the waiting area to the time he/she reaches the waiting area on the opposite side of the road, once when the pedestrian encountered a vehicle, ΔT was calculated. Hitting four points by mouse completed every calculation of ΔT, including the position where the pedestrian stands, the critical collision point between pedestrian and the oncoming vehicle, the vehicle's current position, and the vehicle's position 0.25 s later (the program will automatically show this picture after hitting the former three points).

It is very important to choose the initiative picture for calculation. We use ACDSee 4.0 to scan all pictures and complete this choosing process. For the first vehicle a pedestrian encountered, the initiative picture correspond to the time when the pedestrian happen to arrive at the waiting area, while for other vehicles, the initiative picture correspond to the time when the vehicle in front of them passed over the position where the pedestrian stands.

Finally, 989 samples were collected. Fig. 4 shows the distribution of these samples, which was clustered by whether the pedestrian had passed through the corresponding gap. Fig. 5 shows the relationship between ΔT and the proportion of pedestrian that accept the corresponding gap.

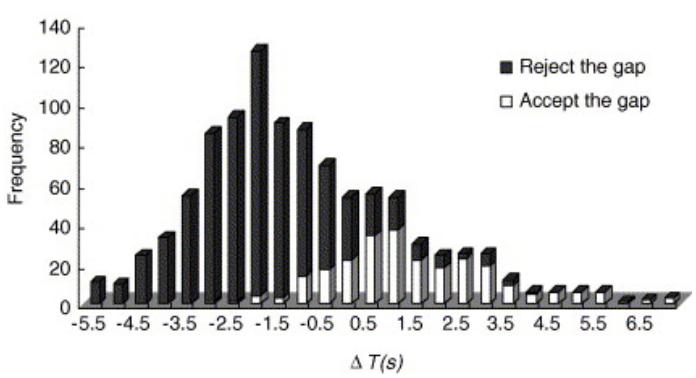

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Fig. 4. Distribution of acquired samples of ΔT.

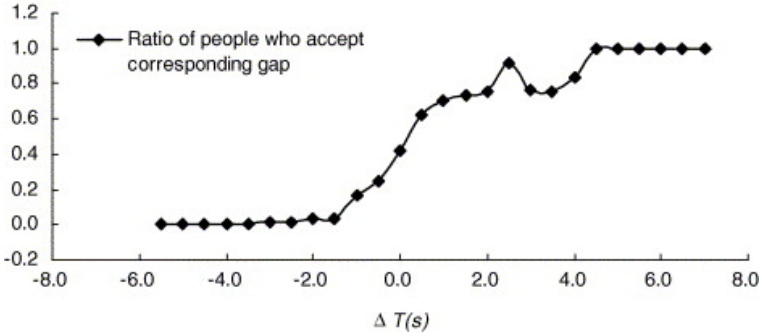

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Fig. 5. Relationship between ΔT and the proportion of pedestrian that accept the corresponding gap.

4. Simulation results

In order to simplify the vehicle model, turning vehicles has been considered as straight going. So the simulating scene becomes a transmeridional road section of an intersection, which has two directions and four lanes, with one pavement on each side. Fig. 6 shows the road map.

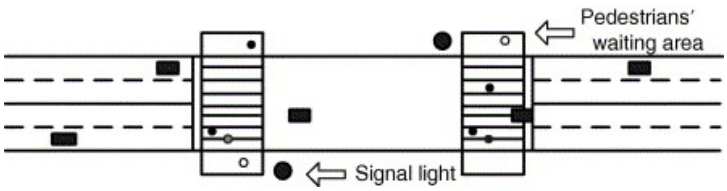

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Fig. 6. Configuration of the crossing in simulation.

Simulation program runs under the Visual C++ 6.0 circumstance. Before its running, we can set whether the police are on duty in dialog box. Participators in the traffic simulation include vehicles and pedestrians. Vehicles in each lane are generated by four car-generators separately. Headways of the vehicles in the same lane are subject to M3 distribution. The simulation platform can count vehicle volume at two pavements, and "pedestrians" could obtain it if they need, together with the police information. In addition, "pedestrian" can also get information about vehicles and other pedestrians to a certain range, then, they will make decisions. We can see pedestrians are marking with three colors: white, black and grey, they represent pedestrians of Type I, Type II, the pedestrians who follow others to violate traffic signal.

From June 2003 to April 2004, our team members recorded the pedestrians' activities at some crossings without police on duty in the city of Xi'an by video camera. We extract information artificially by the recorder explored by ourselves, including vehicle flow rate, pedestrian flow rate, arriving time and leaving time of each pedestrian, the proportion of pedestrians' violation and etc. We use them as reference criterion of the simulation results. In addition, simulation results are extracted by these rules for the same items.

4.1. Pedestrians' generation

Fig. 7 shows comparison of the distribution of pedestrians' arriving intervals between the data extracted from videotape and simulation results. We can see this distribution approximatively subject to M3 distribution (except the original point). Since pedestrians may walk along with their friends or relatives, so the interval, which equals to 0, accounts for a large proportion. So, simply use M3 distribution to describe pedestrians' arriving intervals could not truly represent real situation. We make some changes on the basis of M3 distribution and acquire the simulation results.

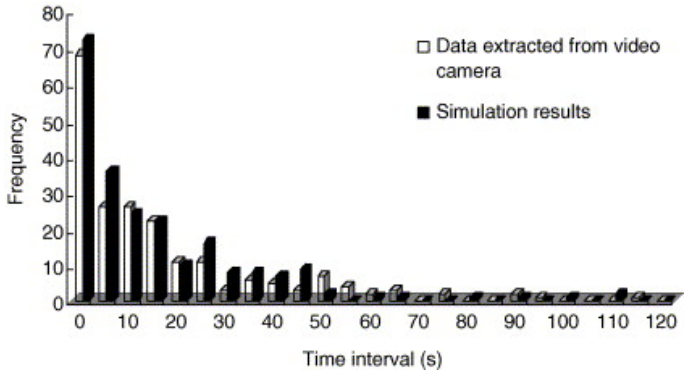

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Fig. 7. Comparison of the time interval distribution of pedestrians' arriving between the data extracted from videotape and simulation results.

4.2. Proportions of red light jumpers and violation followers

Pedestrians who leave the waiting area when the light is red are considered to be red light runners; while the one who leave the waiting area when the light is red after other red light runners within 2 s are considered to be violation followers. Proportion of red light jumpers is the ratio of red light jumpers to people who arrive at the waiting area in red light period. Proportion of violation followers is the ratio of violation followers to red light jumpers.

We use the phases of the traffic lights, vehicle flow rates, and pedestrian volumes as the simulation input parameters. Proportion of pedestrian of Type I(r_1), proportion of violation followers in pedestrian of Type I(r_2), and proportion of pedestrians accept a certain ΔT (r_3) are set according to calibrated parameters. The first group of simulation results is acquired under this parameter setting.

Table 4, Table 5 show the comparison of proportions of red light jumpers and violation followers between data extracted from videotape and simulation results in proportions of red light jumpers and violation followers. We can see that proportion of red light jumpers in simulation results acquired from calibrated parameters is much lower than the data extracted from videotape, while proportion of violation followers is much higher.

Table 4. Comparison of red light runner proportions

Statistics	Data extracted from videotape	Simulation result group 1 (r_1=0.55, r_2=0.69)	Simulation result group 2 (r_1=0.35, r_2=0.69)
Minimum	0.48	0.48	0.58
Maximum	1.00	0.82	0.93
Mean	0.85	0.69	0.82
SD	0.17	0.10	0.10

Table 5. Comparison of violation follower proportions

Statistics	Data extracted from videotape	Simulation result group 1 (r_1=0.55, r_2=0.69)	Simulation result group 2 (r_1=0.35, r_2=0.69)
Minimum	0.00	0.13	0.03
Maximum	0.36	0.33	0.32
Mean	0.18	0.24	0.19
SD	0.11	0.07	0.08

This may due to the degree of confidence of the result of questionnaire. The objects of this investigation are the pedestrians who were willing to complete this questionnaire, and they could not represent all pedestrians. Furthermore, answers provided by some pedestrians are not true response what they do in their daily life. We adjusted parameter r_2 to 0.35 and acquired the second group of simulation results. From Table 4, Table 5, we can see simulation results are more approximate to the data extracted from videotape.

5. Conclusion

We have established a model of pedestrians' road-crossing behavior, which includes the effect of policeman, other pedestrians and vehicles. It represents the behavior of two types of pedestrians (law-obeying ones and opportunistic ones) when facing red light. From the simulation results, violation followers' behavior and phenomenon which vehicles and pedestrians being influenced by each other are represented, and the mixed traffic characteristics are also considered.

Questionnaires and data extracted from videotape are used to calibrate model parameters. However, the simulation results using calibrated parameters deviate from the data extracted from videotape. By adjusting them on the basis of analyzing the occurrence of the deviations, the simulation results agree with field results better.

However, the current work only models pedestrians' decision making of red light running according to the instantaneous states of oncoming vehicles. What pedestrians will do when oncoming vehicle change their states (e.g. accelerate, decelerate, etc) still need to be studied. In addition, the calculation method of ΔT needs to be improved for the turning vehicles so that the model can be more practical for simulating Chinese city traffic.

Recommended articles

References

Blue and Adler, 2001 V.J. Blue, J.L. Adler
Cellular automata microsimulation for modeling bi-directional pedestrian walkways
Transportation Research Part B, 35 (2001), pp. 293-312
View PDF View article View in Scopus ↗ Google Scholar ↗

Das et al., 2005 S. Das, C.F. Manski, M.D. Manuszak
Walk or wait? An empirical analysis of street crossing decisions
Journal of Applied Econometrics, 20 (4) (2005), pp. 529-548
Crossref ↗ View in Scopus ↗ Google Scholar ↗

Fang et al., 2003 W.F. Fang, L.Z. Yang, W.C. Fan
Simulation of bi-direction pedestrian movement using a cellular automata model
Physica A, 321 (2003), pp. 633-640
Google Scholar ↗

Gao et al., 2004 H.B. Gao, H. Chen, L. Shang, Z.G. Lin
The model of pedestrian disturbing in micro-traffic simulation
Technology and Economy in Area of Communications, 4 (2004), pp. 58-60
(in Chinese)
View in Scopus ↗ Google Scholar ↗

Hamed, 2001 M.M. Hamed
Analysis of pedestrian behavior at pedestrian crossings
Safety Science, 38 (2001), pp. 63-82
View PDF View article View in Scopus ↗ Google Scholar ↗

He, 2003 He, R.H., 2003. Applicability of CORSIM and SIMTRAFFIC in Beijing (in Chinese).

Google Scholar ↗

Keegan and Mahony, 2003 O. Keegan, M.O. Mahony
Modifying pedestrian behavior
Transportation Research Part A, 37 (2003), pp. 889-901
View PDF View article View in Scopus ↗ Google Scholar ↗

Liu et al., 2000 Liu, R.H. et al., 2000. A bi-modal microsimulation tool for the assessment of pedestrian delays and traffic management. Paper presented at the 9th International Association of Travel Behavior Research Conference, Gold Coast, July 2000.
Google Scholar ↗

Lyons et al., 2001 G. Lyons, J. Hunt, F. Mcleod
A neural network model for enhanced operation of midblock signaled pedestrian crossings
European Journal of Operational Research, 129 (2001), pp. 346-354
View PDF View article View in Scopus ↗ Google Scholar ↗

Teknomo et al., 2000 Teknomo, K., Takeyama, Y., Inamura, H., 2000. Review on microscopic pedestrian simulation model. In: Proceedings Japan Society of Civil Engineering Conference, Morioka, Japan, March 2000.
Google Scholar ↗

Wan and Rouphail, 2004 Wan, B.H., Rouphail, N.M., 2004. Simulation of pedestrian crossing in roundabout areas using arena, Paper Submitted for publication and presentation at the 83rd TRB Annual Meeting in January 2004.
Google Scholar ↗

Yagil, 2000 D. Yagil
Beliefs, motives and situational factors related to pedestrians' self-reported behavior at signal-controlled crossings
Transportation Research Part F, 3 (2000), pp. 1-13
View PDF View article View in Scopus ↗ Google Scholar ↗

Zhang et al., 2004 J. Zhang, H. Wang, P. Li
Cellular automata modeling of pedestrian's crossing dynamics
Journal of Zhejiang University SCIENCE, 5 (7) (2004), pp. 835-840
View in Scopus ↗ Google Scholar ↗

Cited by (162)

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> …Pedestrian violations include crossing outside of designated markings, failure to yield to vehicles and crossing during a red-light phase. In the literature, violations have been associated with individual characteristics (age and gender) (Guo et al., 2011; Rosenbloom, 2009; Tiwari, Bangdiwala, Saraswat, & Gauray, 2007; Yagil, 2000; Zhuang & Wu, 2011), personal attitudes such as attitude towards subjective norms and social conformity (Evans & Norman, 1998; Moyano Diaz, 2002; Zhou, Horrey, & Yu, 2009), level of pedestrian density, group size (Rosenbloom, 2009; Zhuang & Wu, 2011), vehicular traffic conditions (Wang, Guo, Gao, & Bubb, 2011; Yagil, 2000; Yang, Deng, Wang, Li, & Wang, 2005), waiting time duration (Li & Fernie, 2010; Tiwari et al., 2007; Van Houten et al., 2007), length of the crossing (Cambon de Lavalette et al., 2009; Cinnamon, Schuurman, & Hameed, 2011), weather conditions (Li & Fernie, 2010; Yang et al., 2005), land use (Cinnamon et al., 2011; Zhuang & Wu, 2011) and

trip purpose (Guo et al., 2011). These findings suggest that interventions to address traffic-rule violations should include enforcement and education actions as well as engineering countermeasures, such as the appropriate design of road geometry and traffic controls....

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